

## **Linear Regression Practical Viva Questions & Answers (Based on Python Code)**

### **1. Why did you import pandas?**

Because pandas is used for handling and processing datasets in tabular form.

### **2. Why did you use train\_test\_split?**

To divide the dataset into training and testing data.

### **3. Why did you import LinearRegression?**

To create and train the linear regression model.

### **4. Why did you import r2\_score?**

To evaluate the performance of the model.

### **5. What does this line do?**

```
df = pd.read_csv("housing.csv")
```

It loads the dataset from a CSV file into a DataFrame.

### **6. What is a DataFrame?**

A DataFrame is a table-like data structure in pandas containing rows and columns.

### **7. Why did you separate X and y?**

X contains input features and y contains target/output variable.

Machine learning models learn relationship between inputs and outputs.

### **8. What is MEDV?**

MEDV represents the median value of houses (house price). It is the target variable.

### **9. What does axis=1 mean?**

axis=1 means column operation, so MEDV column is removed from features.

### **10. Why do we split data?**

To train the model on one part and test it on unseen data.

### **11. What does test\_size=0.2 mean?**

20% of the data is used for testing and 80% for training.

### **12. What are X\_train and X\_test?**

- X\_train → Training input data
- X\_test → Testing input data

### **13. What are y\_train and y\_test?**

- y\_train → Training output data
- y\_test → Actual testing output values

### **14. What does this line do?**

```
model = LinearRegression()
```

It creates an object of Linear Regression model.

### **15. What happens during training?**

```
model.fit(X_train, y_train)
```

The model learns relationships between features and house prices.

**16. What is prediction?**

```
model.predict(X_test)
```

The model estimates house prices using testing data.

**17. Why did you create a DataFrame for results?**

To compare actual prices and predicted prices in readable table format.

**18. Why did you use .values?**

To convert pandas series into array format.

**19. What does head(10) do?**

Displays the first 10 rows of the result table.

**20. What is R<sup>2</sup> Score?**

R<sup>2</sup> Score measures how well the model predicts output values.

Closer to 1 means better model.

**21. What does this line do?**

```
round(r2_score(y_test, predictions), 2)
```

Calculates R<sup>2</sup> score and rounds it to 2 decimal places.

**22. What is the formula of Linear Regression?**

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

**23. Is Linear Regression supervised or unsupervised?**

Supervised learning because output labels are already known.

**24. What type of problem is this?**

Regression problem because output is continuous numerical value.

**25. Difference between Classification and Regression?**

Classification predicts categories while Regression predicts numerical values.

**26. Why is Linear Regression suitable here?**

Because house prices generally have linear relationship with features.

**27. What are the limitations of Linear Regression?**

- Assumes linear relationship
- Sensitive to outliers
- Cannot handle complex patterns properly

**28. What will happen if dataset contains missing values?**

The model may give errors or poor predictions, so preprocessing is required.

**29. What is overfitting?**

When model performs very well on training data but poorly on testing data.

**30. What is underfitting?**

When model cannot learn patterns properly from training data.

**31. What libraries from scikit-learn did you use?**

- train\_test\_split

- LinearRegression
- r2\_score

**32. What is machine learning in one line?**

Machine learning enables computers to learn patterns from data and make predictions automatically.

**33. Why is evaluation important?**

To check whether the model performs accurately on unseen data.

**34. Can Linear Regression predict categorical values?**

No. It predicts continuous numerical values only.

**35. Real-life applications of Linear Regression**

- House price prediction
- Sales forecasting
- Stock prediction
- Weather forecasting
- Salary prediction